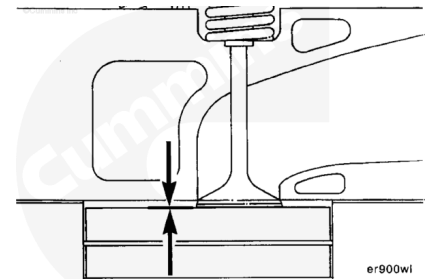


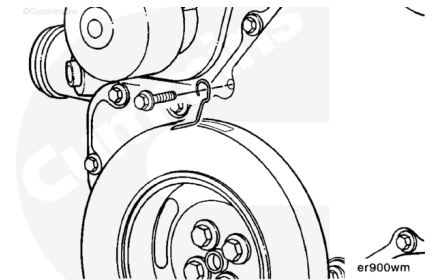
Trainings ▾

Time

Use the Number 1 intake valve to make sure the engine is at top dead center (TDC) on the compression stroke for cylinder Number 1.



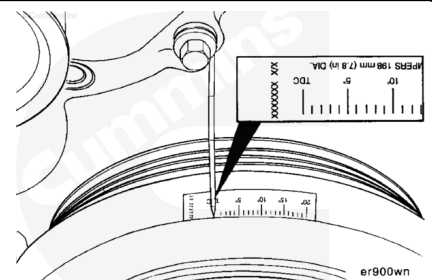
Fabricate a timing mark pointer for the front of the engine. This can be done by forming a piece of wire that can be tightened under one of the gear cover capscrews. Sharpen the wire at the vibration damper end so that it comes to a point for better accuracy.



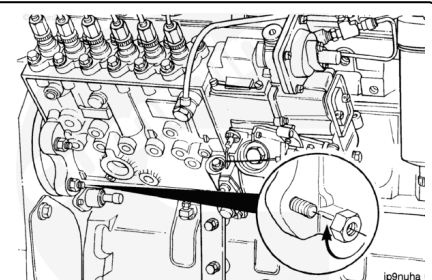
Attach a degree wheel or degree tape to the front of the vibration damper.

Line up the TDC mark with the pointer.

The degree wheel/tape **must** measure to an accuracy of at least 1 degree.

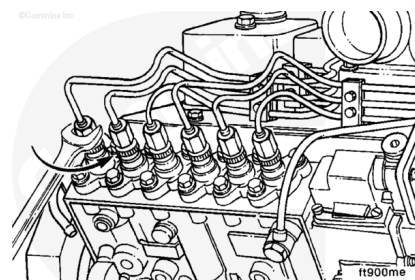


Install the fuel injection pump according to the procedures for a new fuel injection pump installation. If the fuel injection pump is already installed, continue the procedures.



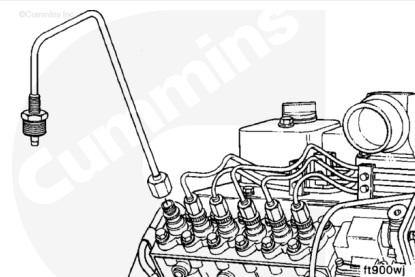
Remove the Number 1 high-pressure fuel line from the fuel injection pump.

Note : Lines 2 through 6 must not be removed or loosened.



⚠ CAUTION ⚠

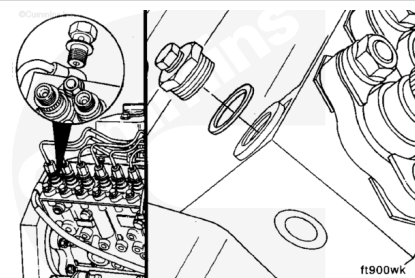
When attaching the fabricated tube, do not bend the Number 1 high-pressure fuel line. This could cause the inside of the fuel line to flake and cause injector failure.



A short length of high-pressure line that is compatible with the fuel lines used on the engine should be bent in a "U" shape and installed onto the delivery valve holder of the fuel injection pump. The line is used to observe when the fuel is or **not** flowing through the delivery valve holder assembly. Place a container under the tube to catch the fuel or drain the fuel back into the spill port pump.

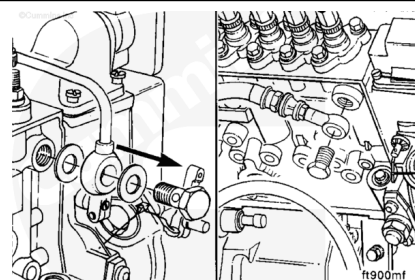
Remove the overflow valve from the fuel injection pump. Install a 14-mm [0.55-in] threaded plug and sealing washer into the fuel return port of the fuel injection pump.

Note : The fuel return port is located on the inboard front side of the fuel injection pump for automotive in-line application and on the outboard front side for most of the industrial applications.

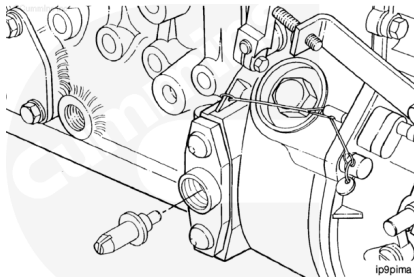


Remove the supply line from the fuel filter head to the fuel injection pump.

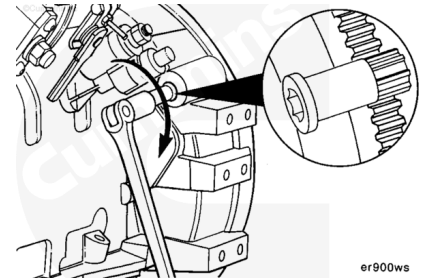
Attach the high-pressure outlet hose from the spill port to the fuel injection pump supply port.



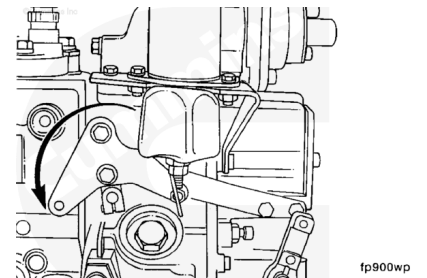
Before continuing, make sure the fuel injection pump lock-timing pin is disengaged.



Rotate the crankshaft **counterclockwise**, as viewed from the front of the engine, to approximately 40 degrees before TDC.



Both the RQV and RQV-K governor require the shutdown lever to be in the full-run position.



⚠ CAUTION ⚠

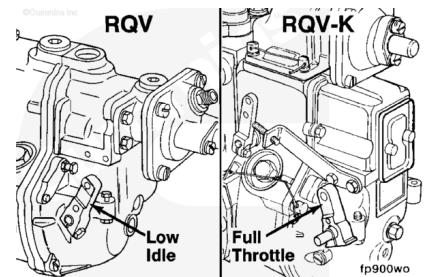
The governor lever must be positioned before pressurizing the fuel injection pump.

Governor Lever Positioning

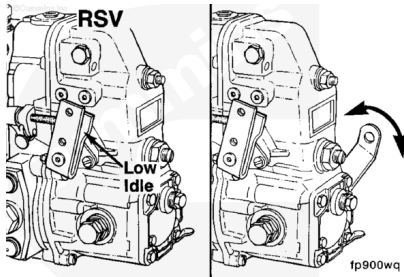
The RQV governor throttle lever **must** be in the low-idle lever position.

The RQV-K governor throttle lever **must** be in the high-idle throttle position on automotive application.

The RQV-K industrial application **must** be in the low-idle position.



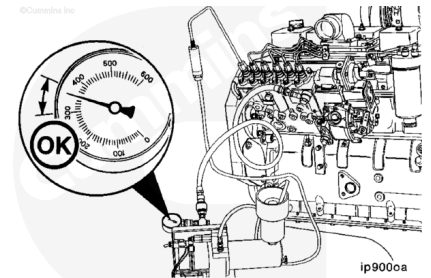
The RSV governor throttle lever **must** be in the low-idle position and the shutdown lever needs to be wired or locked in a suitable fashion to hold the shutdown lever in the 1/2-travel position.



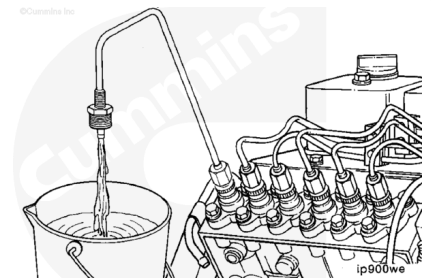
Turn on the spill timing cart pump.

Check the fuel pressure.

The pressure **must** be between 300 psi and 370 psi.

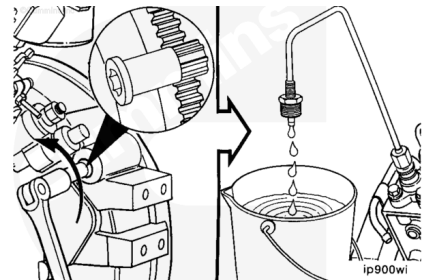


The fuel should be flowing out of the tube attached to the Number 1 cylinder of the fuel injection pump. If the fuel is **not** flowing, recheck the procedures carefully.



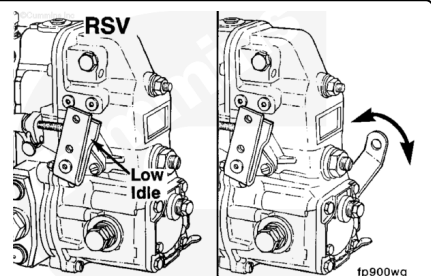
Slowly rotate the crankshaft in the **clockwise** direction, as viewed from the front of the engine, until the fuel flow from the Number 1 cylinder begins to reduce.

The Number 1 plunger element is now approaching "port closure." Continue to rotate the crankshaft slowly until the flow is reduced to a fast drip (more than one drip per second). At the point where the steady stream of flow changes from a solid flow to a fast drip, stop. This is the static timing position of the fuel injection pump.

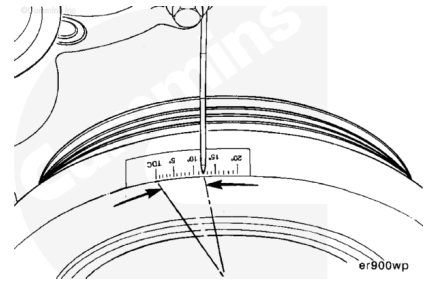


If the flow does **not** slow down to a drip, check the governor position. Also, make sure that the engine is before TDC on the compression stroke.

Turn off the spill port pump.

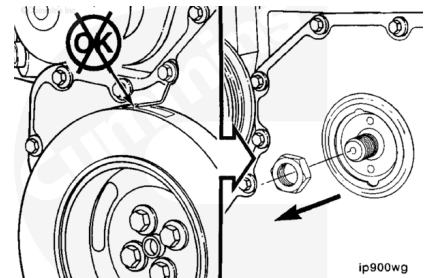


Check the degree wheel on the vibration damper to see what engine degree the timing pointer is indicating. This is spill port static timing. Compare this number to the timing specification for your particular application.



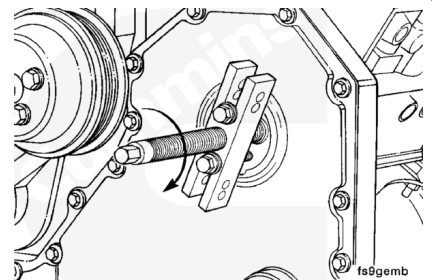
If the fuel injection pump static timing, as measured by the above method, does **not** agree with the specifications you have been given, remove the large nut that fastens the fuel injection pump camshaft to the fuel pump drive gear. If the crankshaft has rotated, turn on the spill port pump, and rotate the crankshaft to find port closure.

Turn off the spill port pump.

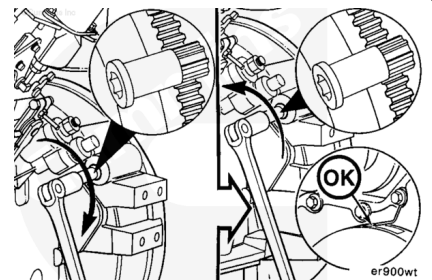


Fuel Pump Gear Puller, Part Number 3163381.

Use a gear puller tool to pull the fuel injection pump drive gear from the fuel injection pump camshaft taper.



Slowly rotate the crankshaft **counterclockwise** about 40 degrees past the desired static timing specification. Slowly rotate the crankshaft in the **clockwise** direction until the timing pointer is indicating the desired static timing.

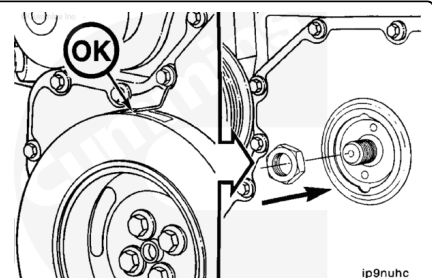


QD Contact Cleaner, Part Number 3824510

Clean the gear taper and pump taper with a nonpetroleum-based cleaner (electrical contact cleaner). Allow surface to dry before installing nut.

Tighten the fuel injection pump drive nut.

Make sure the static timing has **not** changed after the fuel injection drive nut is tightened to the required specification.



Torque Value:

A-Pump

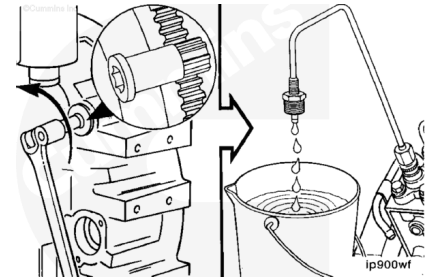
1. 85 n•m [63 ft-lb]

Torque Value:

P3000/P7100

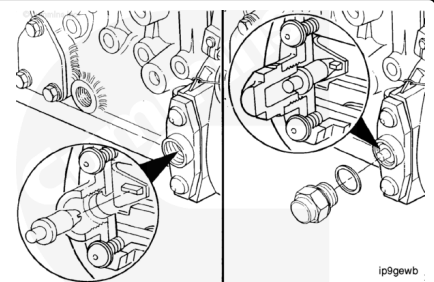
1. 195 n•m [144 ft-lb]

Repeat this procedure as needed until the timing is found to be in agreement with the specification.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



QD Contact Cleaner, Part Number 3824510

Torque Value:

Nippondenso

1. 123 n•m [91 ft-lb]

Torque Value:

Bosch® "A" Pump

1. 85 n•m [63 ft-lb]

Torque Value:

Bosch® In-Line (P3000/P7100) Pump

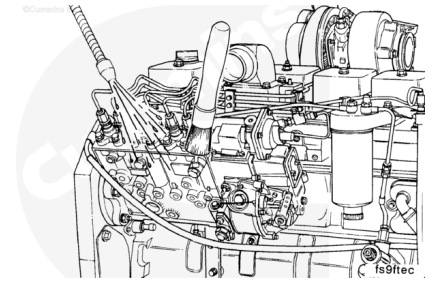
1. 195 n•m [144 ft-lb]

Note : Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with a residue free cleaner (Cummins® Part Number 3824510, or equivalent) by spraying into the gap between the shaft and the gear. Dry the taper surfaces with compressed air. Failure to clean and dry the shaft and gear tapers thoroughly can result in a timing shift to the retarded side after the engine is started and run under load. This will result in low power, smoke, and rough running.

Plunger Lift Timing

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ CAUTION ⚠

Do not allow any dirt, debris, or paint chips to enter the fuel system while it is open. If foreign material of any type is allowed into the pump, lines, or injectors during this process, it could result in an injection pump or fuel injector malfunction.

Note : This timing procedure can only be used on Bosch® P7100 fuel injection pumps.

Plunger Lift Timing Kit, Part Number 3162853

The kit contains the following items:

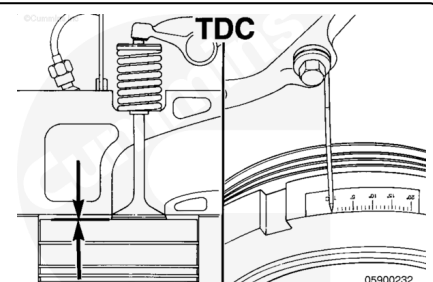
- Dial indicator, Part Number 3824564
- Adapter, Part Number 3824565
- Special socket, Part Number 3824566
- Delivery valve seals, Part Number 3824567.

Step: 1

Thoroughly clean the engine and fuel system before attempting to remove any components. Pay special attention to the top of the fuel injection pump. Use compressed air to remove any water remaining on the fuel pump after the cleaning process.

Step: 2

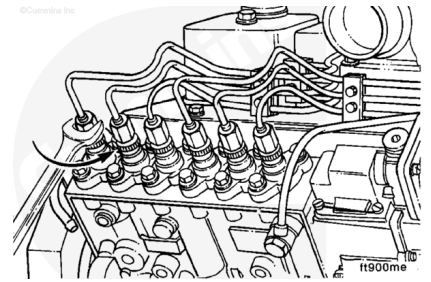
Locate top dead center (TDC) of the compression stroke on cylinder Number 1. The following procedure uses the engine timing pin, but other alternate methods, such as the dropped valve method, can be used.



Step: 3

⚠ CAUTION ⚠

Do not bend the fuel lines. Bending the lines will cause line or injector failure.

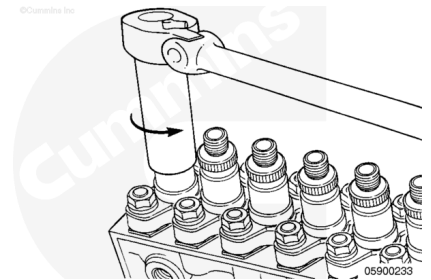


If the injection lines have **not** already been removed, remove the Number 1 injection line.

Step: 4

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



⚠ CAUTION ⚠

Do not loosen the two barrel flange nuts located below the delivery valve holder. Loosening of these nuts will void the fuel injection pump warranty.

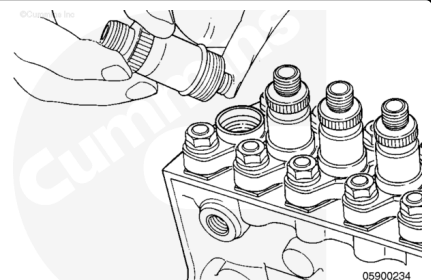
Note : There is an external o-ring on the holder to help prevent debris from getting into the pump; this can create a slight resistance as the holder is loosened.

With the engine at TDC, loosen, but do **not** remove the front (Number 1) delivery valve holder using the special socket provided in the timing kit. Use compressed air to remove any paint chips from around the delivery valve holder. Remove the special socket prior to removing the delivery valve holder from the fuel injection pump.

Step: 5

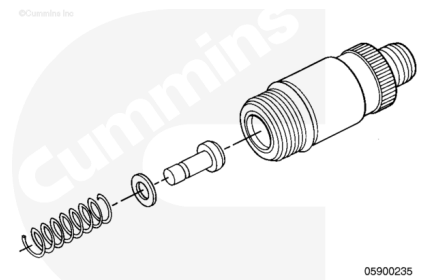
⚠ CAUTION ⚠

Use extreme care when removing the delivery valve holder and delivery valve components. Keep all parts together in the order in which they are removed from the fuel injection pump.



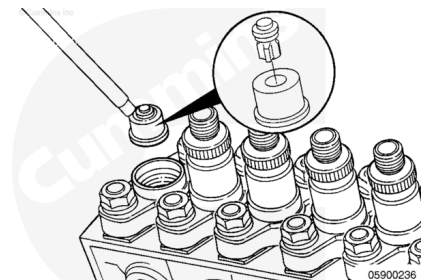
Remove the delivery valve holder by carefully tipping the holder outboard with one hand while using the other hand to hold the spring, fill piece, and any shims from slipping out of the holder.

Place these as an assembly on a clean surface out of the way.



Step: 6

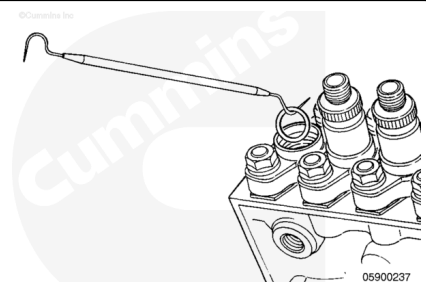
Using a magnet, remove the two-piece delivery valve assembly from the pump. Place these pieces on a clean surface with the delivery valve holder.



Step: 7

⚠ CAUTION ⚠

Do not scratch the top of the plunger/barrel assembly during gasket removal.

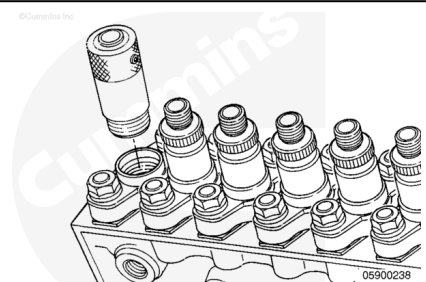


Note : The delivery valve gasket can be either bronze or steel, or on new injection pumps, there is often no gasket.

If the gasket is bronze, use a pick to remove it from the top of the pumping element; if the gasket is steel, use a pick or a magnet. Discard the used delivery valve gasket.

Step: 8

Install the dial indicator from the timing kit in place of the Number 1 delivery valve holder and tighten finger-tight.

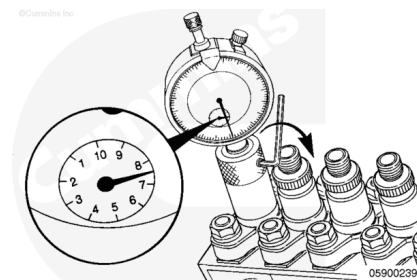


Step: 9

Loosen the set screw on the dial indicator adapter, and install the dial indicator into the adapter. Position the dial indicator to read between 7.0 and 9.0 mm, and lightly tighten the set screw.

Overtightening the set screw will bind the dial indicator.

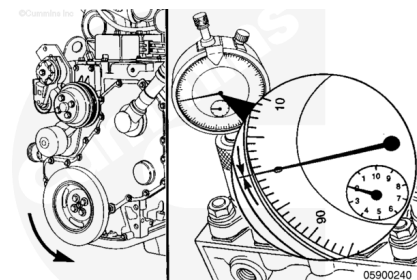
Note : The dial indicator is capable of measuring from 0 to 20.00 mm lift. The small inner dial is marked in increments of 1.00 mm; the large outer dial is marked in increments of 0.01 mm. One revolution of the outer dial is equal to 1.00 mm. The inner dial only indicates 0 to 10.00 mm, but will rotate twice as the indicator goes through the full range.



Step: 10

⚠ CAUTION ⚠

Be sure to disengage the timing pin before rotating the crankshaft to avoid damage to the timing pin.

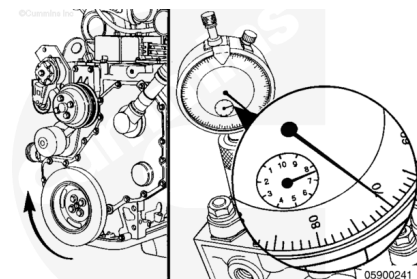


Use a barring tool to rotate the crankshaft in the direction opposite normal direction of crankshaft rotation (**counterclockwise** from front of the engine) 1/4 turn or until the dial indicator reading stops dropping. This is the inner base circle of the fuel injection pump camshaft. Zero "0" the indicator and note the reading on the small inner dial.

Step: 11

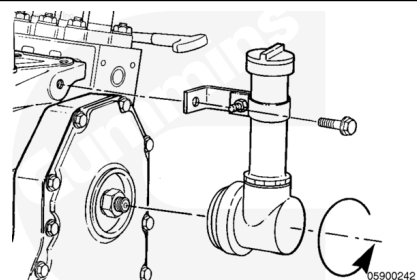
Rotate the crankshaft slowly to TDC in the direction of normal crankshaft rotation (**clockwise** from front of the engine).

The total lift **must** be within the specified lift for that CPL. The attached table lists the lift setting in millimeters (mm) that corresponds to the engine timing specification in degrees.



Step: 12

If a change in injection timing is required, remove the oil filler tube and adapter elbow or fuel injection pump gear access cap from the front of the gear housing.

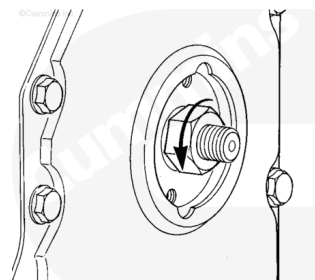


Step: 13

⚠ CAUTION ⚠

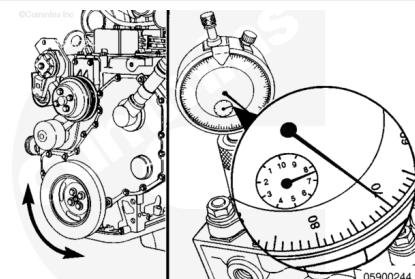
Be sure to disengage the timing pin before rotating the crankshaft to avoid damage to the timing pin.

Loosen the injection pump shaft nut approximately 1/4 turn. The barring tool can be used to keep the crankshaft from rotating.



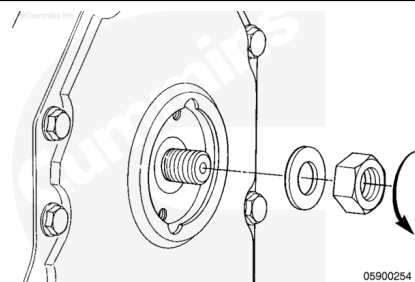
Step: 14

If the actual lift setting is **not** within specification, rotate the crankshaft to obtain the desired dial indicator setting (plunger lift).



⚠ CAUTION ⚠

Do not allow the drive nut or washer to drop inside the engine gear cover. Disassembly of the engine will be required for retrieval.

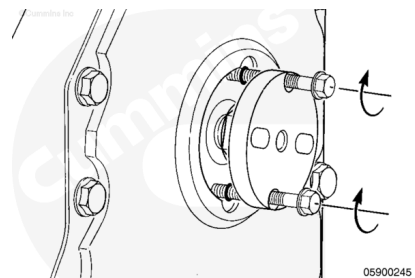


Remove the fuel injection pump driveshaft nut and lock washer.

Service Tip: Use a pointed object such as a metal awl and a magnet, to assist in the removal of the lock washer.

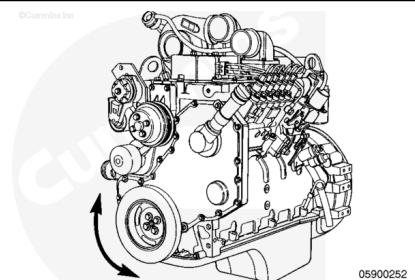
Step: 15

With the fuel injection pump positioned at the correct plunger lift setting, use the gear puller, Part Number 3824469, or equivalent, to pull the injection pump gear off the taper of the injection pump input shaft. Remove the gear puller.



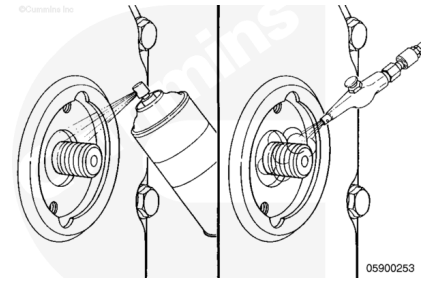
Step: 16

Rotate the crankshaft 20 degrees to 30 degrees opposite the direction of normal crankshaft rotation; then rotate the crankshaft back in the direction of normal crankshaft rotation to TDC. This step removes the backlash from the lower gear train.



Step: 17

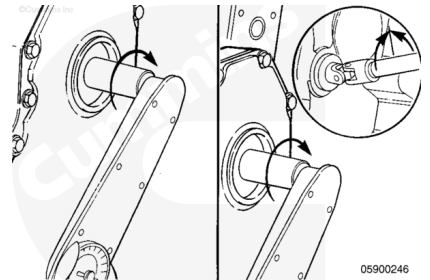
Clean the fuel injection pump shaft and gear tapers with a residue-free cleaner (Cummins® Part Number 3824510, or equivalent) by spraying into the gap between the shaft and gear. Dry the taper surfaces with compressed air. Failure to clean and dry the shaft and gear tapers thoroughly can result in a timing shift to the retarded side after the engine is started and run under load. This will result in low power, smoke, and rough running.



Step: 18

⚠ CAUTION ⚠

Be sure the timing pin is disengaged before the final tightening step to avoid damage to the timing pin.



Install the input shaft lock washer and nut. Use a 2-step tightening process to tighten the fuel injection pump gear nut.

Torque Value: 15 n•m [133 in-lb]

This will seat the shaft taper.

Hold the crankshaft from rotating (using the barring tool or other means) on the final step of the tightening sequence.

Torque Value: 165 n•m [122 ft-lb]

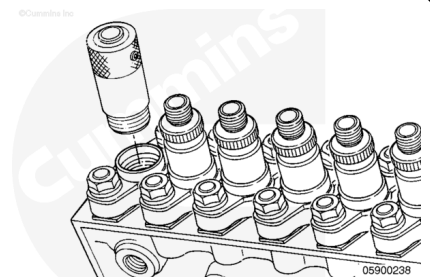
Step: 19

Repeat steps 10 and 11 to verify that the final timing setting is correct.

Step: 20

⚠ CAUTION ⚠

The following steps for installing and tightening the delivery valve and holder must be followed exactly. Improper installation will result in delivery valve damage or leaks.

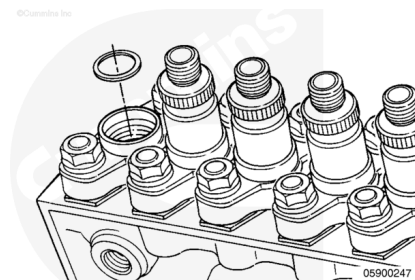


Remove the dial indicator and adapter from the fuel injection pump.

Step: 21

Inspect the fuel injection pump bore for debris. The bore **must** be free of debris.

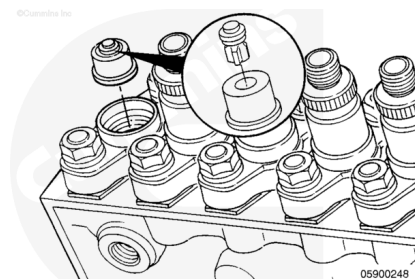
If a gasket was removed from under the delivery valve, install a new delivery valve gasket (Cummins® Part Number 3824567) into the fuel injection pump. Do **not** use a gasket if this is a pump with gasketless delivery valves. See attached list of pumps that have or do **not** have delivery valve gaskets.



Step: 22

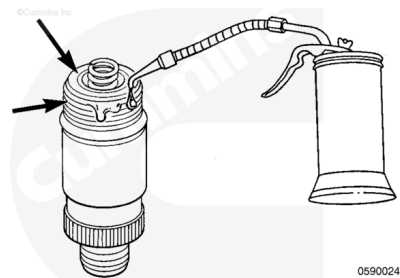
Clean the delivery valve parts with residue-free cleaner (Cummins® Part Number 3824510, or equivalent) prior to reassembly.

Install the delivery valve assembly.



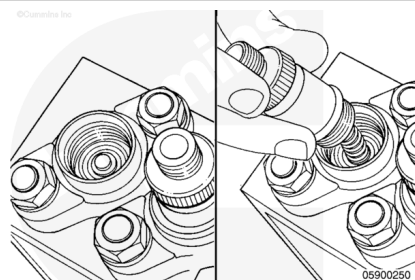
Step: 23

Lubricate the threads and clamping surface of the delivery valve holder with a few drops of SAE 90 hypoid gear oil. Do **not** lubricate the delivery valve gasket or its seating area.



Step: 24

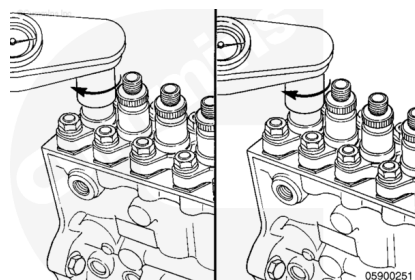
Install the delivery valve holder assembly, taking care **not** to displace the delivery valve spring, fill piece, or any shims.



Step: 25

⚠ WARNING ⚠

Keep hands and body parts away from the high-pressure fuel lines. Fuel coming from the high-pressure fuel lines is under extreme pressure and can cause serious injury by penetrating the skin.



Use step 1 to initially tighten the holder; then, in one motion, use step 2 for the final torque value.

Torque Value:

1. 40 n•m [29 ft-lb]
2. 115 n•m [85 ft-lb]

Install the remaining engine components removed during the TDC location or timing process. Leave the injector side of the high-pressure fuel lines loose to facilitate bleeding the air out of the system.

Crank the engine until fuel is observed at the injectors. Tighten the high-pressure lines at the injector. Start the engine, and vent one line at a time until the engine runs smoothly. Check for leaks.

The following is a list of Bosch® P7100 fuel injection pumps that use delivery valve gaskets. Pumps used in ratings developed after these will use delivery valves without gaskets. If the pump was developed in North America and it is **not** on this list, it will be without gasket. If **not** sure about the type of pump, call Cummins® Care, 1-800-286-6467.

Part Number	
3913340	3921922
3916626	3922424
3916627	3921923
3916628	3921925
3916629	3921970
3917088	3922425
3917089	3922426
3918321	3922427
3919090	3922446
3921769	3922449
3921770	3922471
3921771	3924903
3921772	3925085
3921773	3925086
3921774	3926603
3921775	3927923
3921776	3927924
3921777	3928169
3921918	3928412
3921920	3928606

The following table supplies the lift specifications for the Bosch® P7100 fuel injection pumps **only**. The allowable tolerance range for timing is ± 0.1 mm.

Static Injection Timing (from engine dataplate, degrees BTDC)	987, 1261, 1262, 1263, 1406, 1422, 1582, 1618, 1850, 2118, 2196	1549, 1550, 1551, 1552, 1553, 1815, 1816, 1839, 1914, 1915, 1916, 1959	1863, 1968, 2022, 2174, 2265, 2268, 2308	2023, 2175	1563, 1566, 1569, 1658, 1812, 1905, 1917, 1918, 1920, 1922, 1923, 1949, 1950, 1972, 2034, 2101, 2103, 2104, 2106, 2107, 2108, 2150	1564, 1919, 2011, 2102, 2105, 2158, 2169, 2191
9.5	6.05	5.15	N/A	N/A	5.65	5.6
10	6.15	5.25	N/A	N/A	5.75	5.7
10.5	6.25	5.35	N/A	N/A	5.85	5.8
11	6.35	5.45	4.0	N/A	5.95	5.9
11.5	6.4	5.55	4.05	4.7	6.05	6
12	6.5	5.65	4.15	4.8	6.15	6.05
12.5	6.6	5.7	4.2	4.9	6.25	6.15
13	6.7	5.8	4.3	5	6.35	6.25
13.5	6.8	5.9	4.4	5.1	6.45	6.35
14	N/A	6.0	4.5	5.2	N/A	N/A
14.5	N/A	6.1	4.6	5.3	N/A	N/A
15	N/A	N/A	4.7	5.4	N/A	N/A

The following table supplies the lift specifications for the Marine Midrange engines. The nominal value for control parts list (CPL) 1975 is 22 degrees and CPL 2172 and 2208 is 24 degrees. The allowable range for timing remains at ± 1 degree of nominal.

Control Parts List		
Static Injection	Marine B Series	Marine C Series
Lift Value	CPL(s) 1975, 2208, 8457	CPL 2172
20.0	7.90	7.29
20.5	8.02	7.40
21.0	8.14	7.52
21.5	8.26	7.63
22.0	8.38	7.75
22.5	8.51	7.87
23.0	8.63	7.98
23.5	8.75	8.10
24.0	8.88	8.22
24.5	9.01	8.34
25.0	9.13	8.47
25.5	9.26	8.59
26.0	9.39	8.71